bTBI Synthetic Dataset — FREE SAMPLE Passport

500-row evaluation sample drawn from the 20,000-subject bTBI dataset — same 78-column schema

Field	Value
File	btbi_sample_500.parquet
Rows × Cols	500 × 78 (identical schema to the full dataset — see Appendix A for the full column dictionary)
Selection	True random subset of the full 20,000-row dataset (not a regeneration); fixed selection seed, row indices sorted for reproducibility.
Full dataset	20,000 subjects, version 1.1, SEED 20260926
Sample severity mix	Mild: 405, Moderate: 84, Severe: 7, Penetrating: 4 (counts out of these 500 sampled rows)
Full-dataset severity mix	Severe ~0.5%, Penetrating ~0.4% of all 20,000 subjects. The full dataset carries the exact surveillance marginals; both rare categories can be under-represented or absent in a 500-row sample purely by chance.
Provenance	100% synthetic. No real persons, no real patient data. Parameters: 78.9% verified against published sources, 21.1% anchored estimates (flagged in the full parameter audit), 0 unsourced.
Opening	See PARQUET_HOWTO.txt (included in this delivery) for ready-to-run pandas / R arrow / DuckDB commands.
License	CC BY-NC 4.0 for this free sample (as published on Zenodo) — non-commercial use and redistribution permitted with attribution. The full 20,000-subject dataset requires a commercial license — see sentineldata.com.ua.
Vendor	Sentinel Data — sentineldata.com.ua — ORCID 0009-0004-9749-4867

Intended use

The sample is intended for evaluation of schema, distributions and tooling compatibility — confirm the columns, dtypes and value ranges fit your pipeline before purchasing. Rare categories (Severe 0.5%, Penetrating 0.4% of the full dataset) may be absent or under-represented in 500 rows; the full dataset carries the exact surveillance marginals.

Important caveat

IMPORTANT — CTE risk index: the cte_risk_index column is a hypothesis-generation feature only. A 2017 NIH/NINDS expert panel (Brix et al., J Neurotrauma, doi:10.1089/neu.2017.5222) found no quantitative dose-response evidence linking blast exposure to CTE. Do not use it as a validated risk score.

What ships only with the purchased dataset

This free sample intentionally excludes: the full parameter audit (133 parameters, VERIFIED/ESTIMATE_SOURCED status and anchor source for each), the literature registry (75 sources with DOI/PMID/URL), the automated data-quality report (112 checks), the OMOP CDM v5.4 export, the longitudinal (long-format) events table (540,000 rows), and the deterministic generator script. All of these ship with the purchased dataset — see the full Data Passport for details.

Appendix A — Full column dictionary (78 columns)

Applies identically to btbi_sample_500.parquet and to the full 20,000-row dataset (same schema). Generated from the delivered parquet schema; column order matches the file.

Column	Description
subject_id str · Identifier	Unique synthetic subject identifier. Unit: text. Values/range: BTBI-NNNNNN.
injury_date datetime64[ms] · Context	Synthetic date of the blast injury event. Unit: date. Values/range: 2022-01-02 .. 2024-12-25.
sex str · Demographics	Biological sex. Unit: category. Values/range: Male, Female.
rank_group str · Demographics	Military rank group. Unit: category. Values/range: Enlisted, Officer.
age_years int64 · Demographics	Age at time of injury. Unit: years. Values/range: 18-60.
race_minority_flag bool · Demographics	Flag for minority race group (aggregated, non-identifying). Unit: boolean. Values/range: True/False.
hispanic_flag bool · Demographics	Flag for Hispanic/Latino ethnicity. Unit: boolean. Values/range: True/False.
education_years int64 · Demographics	Years of completed formal education. Unit: years. Values/range: 11-18.
psychiatric_history_flag bool · Demographics	Pre-injury psychiatric history present. Unit: boolean. Values/range: True/False.
blast_mechanism str · Injury context	Mechanism/device causing the blast. Unit: category. Values/range: IED, Mortar, RPG, Rocket, Grenade, Landmine, Aerial bomb.
distance_class str · Injury context	Estimated standoff distance from the blast, banded. Unit: category. Values/range: <=5m, 5-12m, 12-25m, >25m.
helmet_type str · Injury context	Helmet worn at time of injury. Unit: category. Values/range: ACH, ECH, IHPS, LWH.
prior_blast_exposures int64 · Injury context	Count of prior blast exposures (lifetime, before index event). Unit: count. Values/range: 0-109.
lifetime_mtbi_count int64 · Injury context	Lifetime count of mild TBI events including the index event. Unit: count. Values/range: 1-15.
tbi_severity str · Injury context	Clinical severity classification of the index TBI. Unit: category. Values/range: Mild, Moderate, Severe, Penetrating.
gcs_admission int64 · Clinical - acute	Glasgow Coma Scale score at hospital admission. Unit: score 3-15. Values/range: 3-15.
loc_present bool · Clinical - acute	Loss of consciousness present at injury. Unit: boolean. Values/range: True/False.
loc_duration_minutes float64 · Clinical - acute	Duration of loss of consciousness. Extended values represent prolonged coma in severe/penetrating cases. Unit: minutes. Values/range: 0-24987.2.
aoc_present bool · Clinical - acute	Alteration of consciousness present. Unit: boolean. Values/range: True/False.
aoc_duration_hours float64 · Clinical - acute	Duration of alteration of consciousness. Unit: hours. Values/range: 0-789.9.
pta_present bool · Clinical - acute	Post-traumatic amnesia present. Unit: boolean. Values/range: True/False.
pta_duration_hours float64 · Clinical - acute	Duration of post-traumatic amnesia. Unit: hours. Values/range: 0-1837.9.

Column	Description
goat_score_emergence int64 · Clinical - acute	Galveston Orientation and Amnesia Test score at emergence from PTA (-1 = not applicable/not assessed). Unit: score 0-100 or -1. Values/range: -1-100.
ct_abnormal bool · Imaging	Any abnormal finding on head CT. Unit: boolean. Values/range: True/False.
ct_tsah bool · Imaging	Traumatic subarachnoid hemorrhage on CT. Unit: boolean. Values/range: True/False.
ct_ivh bool · Imaging	Intraventricular hemorrhage on CT. Unit: boolean. Values/range: True/False.
ct_mass_lesion bool · Imaging	Mass lesion present on CT. Unit: boolean. Values/range: True/False.
ct_cisterns_abnormal bool · Imaging	Basal cisterns abnormal/compressed on CT. Unit: boolean. Values/range: True/False.
ich_subtype str · Imaging	Dominant intracranial hemorrhage subtype. Unit: category. Values/range: None, EDH, SDH, TSAH, Contusion-IPH.
ct_marshall_grade str · Imaging	Marshall CT Classification grade for diffuse/focal injury. Unit: category I-VI. Values/range: I, II, III, IV, V, VI.
midline_shift_mm float64 · Imaging	Midline shift measured on CT. Unit: mm. Values/range: 0.0-17.5.
midline_shift_gt5mm bool · Imaging	Midline shift greater than 5mm (clinically significant threshold). Unit: boolean. Values/range: True/False.
ct_rotterdam_score int64 · Imaging	Rotterdam CT score (prognostic composite). Unit: score 1-6. Values/range: 1-6.
icd11_primary str · Coding	Primary ICD-11 diagnosis code. Unit: code. Values/range: NA07.0, NA07.02, NA07.05, NA07.06.
icd11_secondary str · Coding	Secondary ICD-11 diagnosis code, if any. Unit: code or None. Values/range: NA07.5, NA07.6, NA07.7, NA07.8, None.
snomed_tbi_code str · Coding	SNOMED CT concept code for the TBI diagnosis. Unit: code. Values/range: 127298000, 127302008.
mace2_baseline_est int64 · Cognitive - acute	Estimated pre-injury baseline MACE2 total score. Unit: score 19-30. Values/range: 19-30.
mace2_total_0h int64 · Cognitive - acute	MACE2 total score at 0 hours (point of injury/initial exam). Unit: score 14-30. Values/range: 14-30.
mace2_total_24h int64 · Cognitive - acute	MACE2 total score at 24 hours. Unit: score 16-30. Values/range: 16-30.
mace2_total_72h int64 · Cognitive - acute	MACE2 total score at 72 hours. Unit: score 18-30. Values/range: 18-30.
mace2_total_7d int64 · Cognitive - acute	MACE2 total score at 7 days. Unit: score 17-30. Values/range: 17-30.
mace2_abnormal_0h bool · Cognitive - acute	MACE2 score flagged abnormal at 0 hours. Unit: boolean. Values/range: True/False.
impact_verbal_memory_baseline_est float64 · Cognitive - ImPACT	Estimated pre-injury baseline ImPACT verbal memory composite. Unit: composite score. Values/range: 51.1-110.3.
impact_verbal_memory_30d float64 · Cognitive - ImPACT	ImPACT verbal memory composite at 30 days. Unit: composite score. Values/range: 47.5-115.1.
impact_verbal_memory_90d float64 · Cognitive - ImPACT	ImPACT verbal memory composite at 90 days. Unit: composite score. Values/range: 51.0-111.3.

Column	Description
impact_visual_memory_baseline_est float64 · Cognitive - ImPACT	Estimated pre-injury baseline ImPACT visual memory composite. Unit: composite score. Values/range: 31.7-101.8.
impact_visual_memory_30d float64 · Cognitive - ImPACT	ImPACT visual memory composite at 30 days. Unit: composite score. Values/range: 21.1-97.2.
impact_visual_memory_90d float64 · Cognitive - ImPACT	ImPACT visual memory composite at 90 days. Unit: composite score. Values/range: 26.3-102.9.
impact_processing_speed_baseline_est float64 · Cognitive - ImPACT	Estimated pre-injury baseline ImPACT processing speed composite. Unit: composite score. Values/range: 9.1-57.0.
impact_processing_speed_30d float64 · Cognitive - ImPACT	ImPACT processing speed composite at 30 days. Unit: composite score. Values/range: 7.9-59.0.
impact_processing_speed_90d float64 · Cognitive - ImPACT	ImPACT processing speed composite at 90 days. Unit: composite score. Values/range: 9.1-61.5.
impact_reaction_time_s_baseline_est float64 · Cognitive - ImPACT	Estimated pre-injury baseline ImPACT reaction time. Unit: seconds. Values/range: 0.28-0.90.
impact_reaction_time_s_30d float64 · Cognitive - ImPACT	ImPACT reaction time at 30 days. Unit: seconds. Values/range: 0.25-0.91.
impact_reaction_time_s_90d float64 · Cognitive - ImPACT	ImPACT reaction time at 90 days. Unit: seconds. Values/range: 0.29-0.94.
voms_symptom_provocation bool · Vestibular/ocular	VOMS testing provoked symptoms. Unit: boolean. Values/range: True/False.
voms_npc_cm float64 · Vestibular/ocular	VOMS near-point-of-convergence distance. Unit: cm. Values/range: 0.2-12.5.
voms_npc_abnormal bool · Vestibular/ocular	Near-point-of-convergence result flagged abnormal (>5cm). Unit: boolean. Values/range: True/False.
voms_vor_positive bool · Vestibular/ocular	Vestibulo-ocular reflex test positive for symptom provocation. Unit: boolean. Values/range: True/False.
headache_chronic bool · Outcomes - symptoms	Chronic post-traumatic headache present. Unit: boolean. Values/range: True/False.
hit6_6mo int64 · Outcomes - symptoms	Headache Impact Test-6 score at 6 months. Unit: score 36-78. Values/range: 36-78.
hit6_12mo int64 · Outcomes - symptoms	Headache Impact Test-6 score at 12 months. Unit: score 36-78. Values/range: 36-76.
pcl5_6mo int64 · Outcomes - mental health	PTSD Checklist for DSM-5 (PCL-5) total score at 6 months. Unit: score 0-80. Values/range: 0-75.
ptsd_screen_positive_6mo bool · Outcomes - mental health	PCL-5 screen positive for probable PTSD at 6 months (standard cutoff). Unit: boolean. Values/range: True/False.
pcl5_12mo int64 · Outcomes - mental health	PCL-5 total score at 12 months. Unit: score 0-80. Values/range: 0-73.
psqi_6mo int64 · Outcomes - sleep	Pittsburgh Sleep Quality Index global score at 6 months. Unit: score 0-21. Values/range: 3-21.

Column	Description
psqi_poor_sleep_6mo bool · Outcomes - sleep	PSQI flags poor sleep quality at 6 months (score >5). Unit: boolean. Values/range: True/False.
psqi_12mo int64 · Outcomes - sleep	Pittsburgh Sleep Quality Index global score at 12 months. Unit: score 0-21. Values/range: 0-21.
phq9_6mo int64 · Outcomes - mental health	Patient Health Questionnaire-9 (depression) score at 6 months. Unit: score 0-27. Values/range: 0-27.
phq9_12mo int64 · Outcomes - mental health	Patient Health Questionnaire-9 score at 12 months. Unit: score 0-27. Values/range: 0-25.
neurocog_residual_90d bool · Outcomes - cognitive	Residual neurocognitive deficit present at 90 days (composite judgment from ImPACT domains). Unit: boolean. Values/range: True/False.
rtd_status_12mo str · Outcomes - functional	Return-to-duty status at 12 months. Unit: category. Values/range: Returned, Not returned, Medical discharge.
label_prolonged_recovery_30d bool · Label	Ground-truth label: recovery prolonged beyond 30 days. Unit: boolean. Values/range: True/False.
label_rtd_failure bool · Label	Ground-truth label: failed to return to duty at 12 months. Unit: boolean. Values/range: True/False.
label_chronic_pcs bool · Label	Ground-truth label: chronic post-concussive syndrome. Unit: boolean. Values/range: True/False.
gose_12mo int64 · Outcomes - functional	Glasgow Outcome Scale Extended score at 12 months. Unit: score 1-8. Values/range: 1-8.
gose_favorable_12mo bool · Outcomes - functional	GOSE score indicates favorable outcome (>=7) at 12 months. Unit: boolean. Values/range: True/False.
cte_exposure_score int64 · Research feature	Cumulative blast-exposure dose proxy (hypothesis-generation feature). No validated dose-response link to CTE exists in the literature (see caveat). Unit: index. Values/range: 1-212.
cte_risk_index int64 · Research feature	Hypothesis-generation CTE risk index. NOT a validated clinical risk score — do not use for diagnosis or risk stratification (per Brix et al. 2017, J Neurotrauma). Unit: index 0-3. Values/range: 0-3.